

5-2d Product Life Cycle

Up to this point, classical theories all paint a *static* picture: If England has an absolute or comparative advantage in textiles (thanks to its favorable weather and soil), it should keep producing textiles. However, this assumption of no change in factor endowments and trade patterns does not always hold true in the real world. While Adam Smith's England, more than 200 years ago, was a major exporter of textiles, today England's textile industry is insignificant. So, what happened? While one may argue that in England weather has changed and soil has become less fertile, it is difficult to believe that weather and soil have changed so much in 200 years, which is a relatively short period for long-run climatic changes. In another example, since the 1980s, the United States turned from being a net exporter to a net importer of personal computers (PCs), while Malaysia transformed itself from being a net importer to a net exporter of PCs—and this change has nothing to do with weather or soil change. Why do patterns of trade in PCs change over time? Classical theories have difficulty answering this intriguing question.

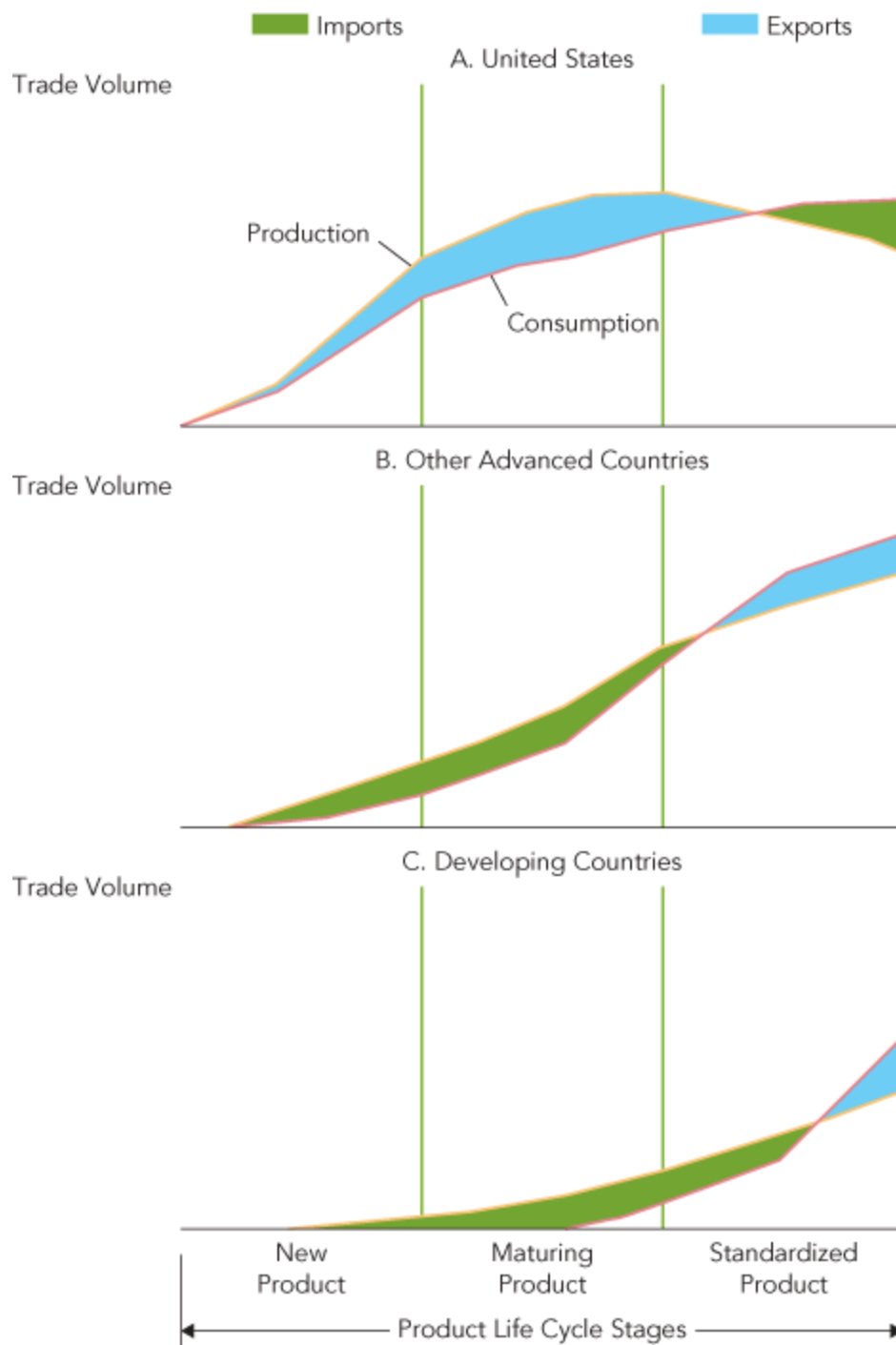
In 1966, American economist Raymond Vernon developed the **product life cycle theory** (A theory that accounts for changes in the patterns of trade over time by focusing on product life cycles.), which is the first *dynamic* theory to account for changes in the patterns of trade over time. Vernon divided the world into three categories:

- (1) lead innovation nation (which, according to him, is typically the United States),
- (2) other developed nations, and
- (3) developing nations.

Further, every product has three life cycle stages: new, maturing, and standardized. Shown in **Figure 5.4**, in the first stage, production of a new product (such as a TV) that commands a price premium will concentrate in the United States, which exports to other developed nations. In the second, maturing stage, demand and ability to produce grow in other developed nations (such as Australia and Italy), so it is now worthwhile to produce there. In the third stage, the previously new product is standardized (or commoditized). Thus, much production will now move to low-cost developing nations, which export to developed nations. In other words, comparative advantage may change over time.

Figure 5.4

Theory of Product Life Cycles



While this theory was first proposed in the 1960s, some later events (such as the migration of PC production) have supported its prediction. However, this theory has been criticized on two accounts. First, it assumes that the United States will always be the lead innovation nation for new products. This may be increasingly invalid. For example, the fanciest cell (mobile) phones are now routinely pioneered in Asia and Europe. Second, this theory assumes a stage-by-stage migration of production, taking at least several years (if not decades). In reality, however, an increasing number of firms now *simultaneously* launch new products (such as iPads) around the globe.

Printed By: Mike Benton (mbento6@wgu.edu)
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